

Statistical Analysis

3rd Year BE

By

Mysore University School of Engineering

University of Mysore

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Statistical Analysis (21AI62)

Semester VI

No. of Lecture hour/Week	3	CIE Marks	50
No. of Tutorial hours/week	0	SEE Marks	50
Total No. of Lecture hours	50	Exam Hours	03
L: T:P	3:0:2	Credits	03

Modules	Course Content	Teaching Hours
Module 1	Data and Representation: Introduction to Statistics, Collection of data, classification and tabulation of data. Types of data: Primary data, Secondary data. Presentation of data Diagrammatic and Graphical Representation: Histogram, frequency curve, frequency polygon, Ogive curves, stem and leaf chart.	10 Hours
Module 2	Measures of Central Tendency: Mean (A.M.) Definition, Mode, Median. Partition Values: Quartiles, Deciles and Percentiles, Box Plot, Percentile ranks. Means of transformed data, Geometric Mean (G.M.) Definition, Harmonic Mean (H.M.), Weighted Mean: Weighted A.M., G.M. and H.M.	10 Hours
Module 3	Dispersion Arithmetic: Range, Mean deviation Mean square deviation, Variance and Standard Deviation, Combined variance (derivation for 2 groups), Combined standard deviation.	10 Hours
Module 4	Correlation and Regression: Bivariate normal distribution, types, importance, methods of measuring correlation-scatter diagram, Karl Pearson's Coefficient of Correlation and Spearman's rank Correlation. Regression lines, Difference between regression and correlation, uses of Regression	10 Hours
Module 5	Sampling theory and tests of significance: Methods of sampling (Description only): Simple random sampling with and without replacement (SRSWR and SRWOR) stratified random sampling, systematic sampling. Tests of significance – z, t, chi-square and F.	10 Hours

Course Outcome:

At the end of the course the student will be able to:

CO1: Differentiate between primary and secondary data to enhance data reliability.

CO2: Interpret quartiles, deciles, and percentiles to understand data distribution.

CO3: Compute standard deviation to assess the variability within datasets.

CO4: Apply regression lines for predictive analysis based on correlation coefficients.

CO5: Apply chi-square test to evaluate the significance of categorical data.

Reference Books:

1. Statistics for Business and Economics, by Paul Newbold, William L. Carlson, and Betty Thorne;
2. Statistics, by David Freedman, Robert Pisani, and Roger Purves;
3. Statistical Methods for Research Workers by Ronald A. Fisher;
4. Introduction to the Practice of Statistics by David S. Moore, George P. McCabe, and Bruce A. Craig.

MODULE 1

DATA AND REPRESENTATION**Table of Content**

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1.1 Statistics

The term "statistics" is derived from the Latin word "status," the Italian word "statista," or the German word "statistika." Each of these terms refers to the political state. Initially, statistics were used to gather information on the state's citizens' income, health, illiteracy, and wealth, among other things.

However, statistics has evolved into an essential science with practical applications in a variety of sectors of everyday life.

1.1.1 The Role of Statistics and the Data Analysis Process.

Statistics is a scientific field that teaches us how to evaluate and interpret statistics, providing valuable tools for learning about the world around us. Its vast application in a variety of sectors has led to the understanding that statistical literacy is an essential component of a well-rounded educational curriculum. Statistics teaches us how to make sound judgments in the face of uncertainty and variability.

This chapter investigates the nature and use of variability in statistical situations, defines essential vocabulary, and includes examples of graphical representations for data summation.

1.1.2 Why Statistics?

There is an old phrase that "without data, you are just another person with an opinion." Although stories and coincidences can provide interesting stories, they should not be used to make important decisions.

For instance, a friend's experience with apricots for joint pain does not guarantee the effectiveness of a treatment for arthritis. To make informed recommendations, relevant data should be considered, allowing for an investigation of the treatment's effectiveness.

Raw observations alone are data, but they are not information or knowledge. Data raises questions, such as:

- *What is the most common or expected observation?*
- *What are the limits on the observations?*
- *What does the data look like?*

Although they appear simple, these questions must be answered in order to turn raw observations into information that we can use and share. Beyond raw data, we may design experiments in order to collect observations.

From these experimental results we may have more sophisticated questions, such as:

- *What variables are most relevant?*
- *What is the difference in an outcome between two experiments?*
- *Are the differences real or the result of noise in the data?*

To effectively study statistics, we should be able to:

1. *Extract information from tables, charts, and graphs.*
2. *Follow numerical arguments.*
3. *Understand the fundamentals of data collection, summarization, and analysis to draw statistical conclusions.*

Throughout your personal and professional life, understanding and using data to make decisions is essential.

To do this, you must be able to:

1. *Determine whether existing data is sufficient or if additional information is needed.*
2. *Collect additional information in a reasonable and thoughtful manner, if necessary.*
3. *Summarize the available data in a useful and informative way.*
4. *Analyze the available data.*
5. *Draw conclusions, make decisions, and assess the risk of an incorrect decision.*

1.1.3 What is Statistics?

Statistics is a sub field of mathematics. It refers to a collection of methods for working with data and using data to answer questions.

Statistics is the art of making numerical conjectures about puzzling questions. [...]

The methods were developed over several hundred years by people who were looking for answers to their questions.

| *Page xiii, Statistics, Fourth Edition, 2007.*

Statistics allow researchers to collect information, or data, from a large number of people and then summarize their typical experience.

[...] Statistics are also used to reach conclusions about general differences between groups. [...] Statistics can also be used to see if scores on two variables are related and to make predictions.

| *Pages ix-x, Statistics in Plain English, Third Edition, 2010.*

1.1.4 Characteristics of Statistics

1. **Aggregate of Facts:** Statistics are collections of data.
2. **Numerically Expressed:** Statistics are represented with numbers.
3. **Affected by Multiple Causes:** Various factors influence statistics significantly.
4. **Accuracy:** Statistics are measured or estimated with a reasonable degree of precision.
5. **Systematically Collected:** Data is gathered in an organized way.
6. **Purpose-Driven:** Data is collected with a specific goal in mind.
7. **Interrelated:** Statistics should be considered in relation to other data.

Important Note: *Not all numerical data are statistics. For data to be considered statistics, it must meet the above criteria. Thus, "all statistics are numerical statements of facts, but not all numerical statements of facts are statistics."*

1.1.5 Important Terminology in Statistics

1. Population:

- A well-defined set or group of all objects for a particular study.
- Examples: Persons, plants, books, fishes in ponds, shops, etc.
- Elements: Specific items within the population, like plants in a field, fishes in a pond, unemployed persons in a country, books in a library.
- **Finite Population:** Limited number of elements (e.g., students in a school).
- **Infinite Population:** Unlimited number of elements (less common).

2. Sample:

- A part of the population selected by some sampling procedure.
- **Sampling:** The process of selecting a sample.
- **Sample Size:** The number of objects in the sample.
- **Purpose:** To represent the population.
- **Example:** Studying the weight of fishes in a pond:
 - **Complete Enumeration (Census):** Measuring the weight of all fishes (3,000 fishes).
 - **Sample Survey:** Measuring the weight of a selected group of fishes.

3. Variable:

- A property of objects that varies from object to object and is numerically expressible.
- **Example:** Marks in Mathematics of students in a class can be expressed in terms of the marks obtained.

4. Attribute:

A property or characteristic of objects that is not expressible quantitatively in numbers.

Example: Qualitative characteristics like smoking, color, honesty.

5. Definition and Scope:

- **Science and Art:** Statistics involves both scientific principles and creative application.

- **Principles and Methods:** It encompasses the collection, presentation, analysis, interpretation, and forecasting of data.
- **Decision Making:** Statistical facts, influenced by multiple factors and applicable to various fields, help in making informed decisions under uncertainty.

1.1.6 Stages in a Statistical Investigation



Fig1: Stages in a Statistical Investigation.

1. Collection:

- The primary step in a statistical study.
- **Importance:** Careful collection is crucial as faulty data lead to unreliable conclusions.
- **Sources:** Data can be obtained from existing published or unpublished sources, or collected first-hand by the investigator.
- **Challenges:** First-hand data collection is one of the most difficult and important tasks for a statistician.

2. Organization:

- **Purpose:** To arrange a large mass of figures from a survey into an organized form.
- **Steps in Organization:**
 - **Editing:** Carefully review collected data to correct or adjust omissions, inconsistencies, irrelevant answers, and wrong computations.
 - **Classification:** Arrange data according to common characteristics of the items.
 - **Tabulation:** Arrange data in columns and rows for clarity.
- **Outcome:** Proper organization highlights desired information and avoids undesirable information.

3. Presentation:

- **Need:** Organized data alone cannot influence a layman.
- **Methods:** Present data using tables, diagrams, and graphs.
- **Benefit:** These visual aids make facts easy to understand.

4. Analysis:

- **Focus:** Methods used to analyze presented data, usually in tabular form.
- **Tools:** Averages, correlation, regression, and other statistical tools.

5. Interpretation:

- **Challenge:** Requires a high degree of skill and experience.
- **Importance:** Decisions are made based on the conclusions drawn from data interpretation.

1.1.7 Types of Statistics

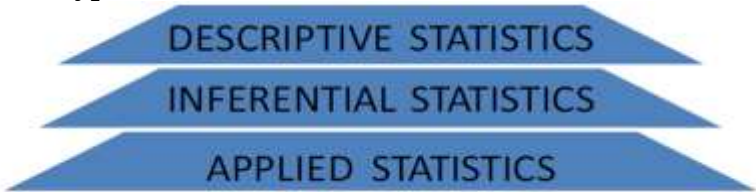


Fig 2: Types of Statistics.

1. **Descriptive Statistics:** Descriptive statistics refer to methods for summarizing raw observations into information that we can understand and share.
Purpose: Summarizes raw data into understandable information.
Methods:
 - **Statistical Values:** Calculating mean, median, variance, and standard deviation.
 - **Graphical Methods:** Using charts and graphics to visualize data, showing distribution and relationships between variables.**Components:**
 - **Central Tendency:** Measures like mean and median.
 - **Dispersion:** Measures like variance and standard deviation.
 - **Visualization:** Graphs and charts for qualitative understanding.
2. **Inferential Statistics:** Inferential statistics is a fancy name for methods that aid in quantifying properties of the domain or population from a smaller set of obtained observations called a sample. These are often referred to as tools for statistical hypothesis testing, where the base assumption of a test is called the null hypothesis.
Purpose: Quantifies properties of a population based on a sample.

Methods:

- **Estimation:** Inferring population parameters like expected value and spread.
- **Hypothesis Testing:** Tools for testing assumptions (null hypothesis) and quantifying the likelihood of observations.

Applications:

- **Probability Theory:** Estimating the likelihood of outcomes.
- **Sampling Techniques:** Drawing conclusions about a population from sample data.

3. **Applied Statistics** It involves application of statistical methods and techniques to the problems and actual facts.

Purpose: Application of statistical methods to real-world problems.

Subdivisions:

- **Descriptive Applied Statistics:** Provides descriptive information about known data (e.g., price index numbers, vital statistics).
- **Scientific Applied Statistics:** Uses statistical methods to formulate and verify scientific laws (e.g., law of demand, quantitative theory of money).

Examples: Statistics related to national income, industrial and agricultural production, population, and prices.

1.1.8 Importance of Statistics

- **Prerequisite for Machine Learning:** Understanding statistics is essential for applied machine learning.
- **Transforming Observations:** Statistics helps convert raw data into useful information.
- **Summarizing Data:** Developed over hundreds of years, statistical tools help summarize data and quantify properties of a domain based on samples.

Statistics plays a crucial role in various aspects of life, from government planning to business operations.

Its significance is evident in the following areas:**1. Importance to the State or Government:**

- Governments rely on statistics to formulate plans and policies for the welfare of their citizens.
- Statistical data provide the basis for government decisions, such as the framing of five-year plans.

2. Importance in Human Behavior:

- Statistical methods like averages and correlations help understand human activities and behavior.

- For example, individuals collect data on prices from different shops to determine average prices and price ranges before making purchases.

3. Importance in Economics:

- Economics is becoming increasingly statistical in its methods.
- Statistical principles underpin various economic concepts, such as the law of demand and elasticity of demand.
- Statistics are used in consumption, production, and distribution sectors to analyze capital investment, output, and national income.

4. Importance in Planning:

- Planning for resource utilization and national growth relies heavily on statistical information.
- Organizations like the National Sample Survey Organization (NSSO) and Central Statistical Organization (CSO) in India provide vital statistical data for planning purposes.

5. Importance in Business:

- Businesses cannot thrive without the use of statistics in the modern era.
- Statistical techniques aid in marketing strategies, quality control, banking decisions, accounting practices, and research and development.
- Examples include analyzing market data for effective marketing strategies, using control charts for quality control, and making investment decisions based on statistical analysis.

1.1.9 Limitations of Statistics

Statistics, while a valuable research tool, is subject to several limitations that must be acknowledged

1. Qualitative Facts:

- Statistics focuses on numerical facts and does not study qualitative phenomena directly.
- Qualities such as morality or affection cannot be quantified and fall outside the scope of statistics.

2. Individual Study:

- Statistics deals with groups and averages, not individual cases.
- Individual causes or outliers may not be adequately addressed through statistical analysis.

3. Average Results:

- Statistical laws are true only on average and are probabilistic, not deterministic.

- While statistical conclusions provide insights, they do not guarantee specific outcomes for every case.

4. Lack of Complete Accuracy:

- Statistical data are approximations or estimates rather than precise measurements.
- Census or sample data may not fully represent the true population, leading to inaccuracies.

5. Potential for Misuse:

- Statistics can be manipulated or distorted if handled by inexperienced individuals.
- Expertise is necessary to ensure proper handling and interpretation of statistical data.

6. One of Many Methods:

- Statistics is just one approach to studying phenomena and may not be suitable for every research problem.
- Other methods should be considered and integrated with statistical analysis for comprehensive understanding.

7. Potential for Misleading Results:

- Statistical results may be misinterpreted or lead to misleading conclusions if not properly contextualized.
- Without careful consideration of factors beyond statistical data, conclusions may be erroneous.

1.1.10 Distrust of Statistics

While statistics is a valuable science for practical purposes, its misuse and misinterpretation can lead to distrust in its findings. Here are two contrasting opinions on the issue

1st Opinion: Trust in Statistical Conclusions

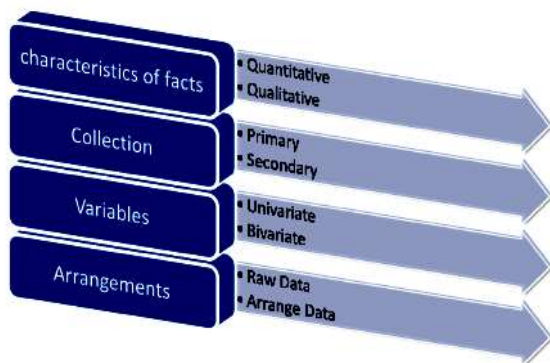
- Statisticians trust in the rigor of statistical methods used for data collection, editing, analysis, and interpretation.
- They believe that statistical conclusions are reliable and accurate, as they are based on sound principles.
- Phrases like "Figures don't lie" or "Figures can prove nothing" reflect this viewpoint.

2nd Opinion: Suspicion and Distrust

- Many people view statistics with suspicion, considering it prone to manipulation and misrepresentation.
- Statistics are sometimes condemned as being deceptive, with phrases like "Figures are a tissue of falsehood" or "There are lies, damned lies, and statistics."
- Some believe that statistics can be used to distort truth, with phrases like "Statistics can prove anything" or "Statistics are like clay that can be molded to fit any narrative."

1.2 Types of Data

Data form the fundamental building blocks of statistical investigations, encompassing various characteristics and arrangements. Here are the types of data:



1.2.1 Based on the Characteristics of facts:

1. Quantitative Data or Numerical Data:

- These data can be directly measured, such as age, income, production, or marks.
- Quantitative data consist of variables, which may be discrete or continuous.
 - **Discrete Variables:** Values are individually distinct and discontinuous, such as the number of students in a class or books on a library shelf.
 - **Continuous Variables:** Values form an uninterrupted range, such as age, weight, or height.

Discrete data or Continuous data?

- No. of accidents on each day in a month (Discrete)
- Lengths of 1,000 bolts produced in a factory (Continuous)
- Speed of an automobile in kilometer per hour (Continuous)
- No. of books on a library shelf (Discrete)

2. Qualitative Data or Categorical Data:

- These data pertain to facts that cannot be directly measured but are counted or categorized based on attributes.
- Examples include attributes like literate, illiterate, unemployed, or honest.
- Qualitative data may further be classified into two categories.

1.2.2 Based on the Source of Collection:

1. **Primary Data:** Collected firsthand for specific purposes, using methods like direct personal investigation, indirect investigations, correspondents, mailed questionnaires, and schedules.

Definition: Data originally collected for the first time for a specific survey or study.

Example: Collecting data on tea or coffee drinking habits in a village by an investigator.

Methods of Collecting Primary Data:

1. **Direct Personal Investigation:** The investigator personally collects data from respondents.
 - a. **Advantage:** Most reliable.
 - b. **Disadvantage:** Results may be influenced by the investigator's personal bias.
2. **Indirect Investigations:** Information is obtained indirectly from a third person who knows the facts about the respondent.

Usage: Commonly used by government-appointed commissions.
3. **Through Correspondents:** Correspondents or agents are appointed to collect and transmit information from various places to the central office.

Usage: Widely used by newspaper offices.
4. **By Mailed Questionnaire:** A structured questionnaire is mailed to respondents, who fill it out and send it back.

Limitation: Effective only with literate respondents.
5. **Through Schedules:** Field workers visit respondents with a schedule of questions and collect answers in their handwriting.

Usage: Useful for illiterate respondents.
2. **Secondary Data:** Collected by someone else and used by other entities, available in published forms (reports, publications) and unpublished forms (internal reports, records). Data that has already been collected by others and is used by a different entity.

Example: Data collected by the government for a census is primary for the government but secondary for researchers using it later.

Sources of Secondary Data:

- **Government Publications:** Census data, economic surveys, health reports.
- **International Organizations:** World Bank, United Nations, International Monetary Fund.
- **Academic Research:** Published papers, theses, dissertations.
- **Business Records:** Annual reports, sales records, market research data.

Types of Secondary Data.

- i. **Published Sources:** Published reports in newspapers, Reserve Bank of India (RBI) publications, and periodicals.
 - ✓ Publications from trade associations.
 - ✓ Financial data in annual reports.
 - ✓ Data available in Securities and Exchange Board of India (SEBI) publications.
 - ✓ Official publications from government bodies.
 - ✓ Publications of international bodies such as the United Nations (UNO), World Bank, etc.
 - ✓ Other published sources.
- ii. **Unpublished Sources:**
 - ✓ Internal reports from government departments.
 - ✓ Records maintained by institutions.
 - ✓ Research reports prepared by university students.

1.2.3 On the Basis of Variables:

1. **Univariate Data:** Frequencies determined based on one variable, such as the number of workers categorized by wages or age.
2. **Bivariate Data:** Data analyzed based on two variables simultaneously, typically presented in a two-way frequency table.

1.2.4 On the Basis of Arrangements:

1. **Raw Data:** Unprocessed data that has not undergone statistical methods.
2. **Arrange Data:** Processed data that has been organized, summarized, classified, and tabulated properly.
Terms like 'Data Point' and 'Data Set' differentiate between individual facts and aggregates of facts. For instance, production

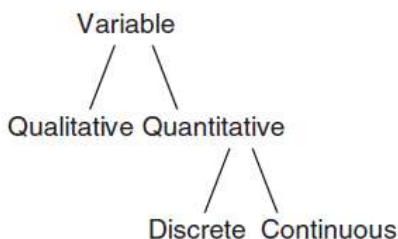
data for ten years constitute a 'Data Set,' while figures for one year represent a 'Data Point.'

1.3 Collection of Data

A data set is composed of information from a set of units. Information from a unit is known as an **observation**. An observation consists of one or more pieces of information about the unit; these are called **variables**.

Variables that yield nonnumerical information are called **qualitative** variables. Qualitative variables are often referred to as categorical variables. Those that yield numerical measurements are called **quantitative** variables. Quantitative variables can be further classified as discrete or continuous.

The diagram below summarizes these definitions:



A **discrete variable** can assume only a countable number of values. Typically, discrete variables are frequencies of observations having specific characteristics, but all discrete variables are not necessarily frequencies.

A **continuous variable** is one that can take any one of an uncountable number of values in an interval. Continuous variables are usually measured on a scale and, although they may appear discrete due to imprecise measurement, they can conceptually take any value in an interval and cannot therefore be enumerated.

In the field of statistical quality control, the term **variable data** is used when referring to data obtained on a continuous variable and **attribute data** when referring to data obtained on a discrete variable.

Getting **information** on various things around us has become a way of life. Information itself is a major source of knowledge. Without information it is difficult to take decisions.

- With development of science and technology the sources of information have increased and become accessible as well.
- Books, Newspapers, magazines, telephone, television, internet and mobile phones etc. are all medium of providing information of various kinds.

Information is both **qualitative** and **quantitative** in nature. Good, bad, ugly, beautiful, responsible, noble, handsome, educated etc are terms used to describe persons, can be said to be qualitative in nature. Information on income, expenditure, savings, rate of growth, height, weight, marks secured, population, food production, etc are given in quantitative or numerical terms. In the study of economics quantitative information's are mostly used for analysis.

1.3.1 Nature of Data

To understand the nature of data, it is important to recognize the different contexts in which data can be collected and analyzed. Data can be categorized based on the dimension it represents, such as time, geographical location, or a combination of both.

Here are the three primary types of data:

1. **Time Series Data:** Time series data consists of a sequence of numerical values collected at different points in time. This type of data helps in analyzing trends, patterns, and cycles over a period. Time series data can be collected at regular intervals (e.g., daily, monthly, yearly) or irregular intervals.

Example: Monthly sales revenue of a company over the past five years.

Characteristics:

- **Temporal Order:** The data points are ordered chronologically.
- **Interval Consistency:** Can be regular (e.g., daily temperature readings) or irregular (e.g., timestamps of events).

Applications:

- Forecasting future trends (e.g., stock prices, economic indicators).
 - Identifying seasonal patterns (e.g., sales peak during holidays).
 - Analyzing time-dependent phenomena (e.g., climate change).
2. **Spatial Data:** Spatial data, also known as geospatial data, represents information about the physical location and shape of objects, as well as their relationships in space. This type of data is essential for mapping and geographic analysis.
- Example:** Locations of coffee shops in a city.

Characteristics:

- **Geographical Coordinates:** Data points are associated with specific locations (latitude and longitude).
- **Spatial Relationships:** Includes information about proximity, distance, and spatial patterns.

Applications:

- Urban planning and management (e.g., distribution of public services).
- Environmental monitoring (e.g., mapping deforestation).
- Navigation and transportation (e.g., GPS systems).

3. **Spatio-Temporal Data:** Spatio-temporal data combines both time and space dimensions. It involves data that varies across both time and geographical locations, providing a comprehensive view of how phenomena evolve over time and space.

Example: Daily air quality measurements across different cities over a year.

Characteristics:

- **Dual Dimensions:** Incorporates both temporal and spatial attributes.
- **Complex Analysis:** Requires methods to analyze changes over both time and location.

Applications:

- Tracking the spread of diseases (e.g., COVID-19 cases over time and across regions).
- Climate studies (e.g., changes in temperature patterns across different areas over decades).
- Transportation analysis (e.g., traffic flow changes over time in various parts of a city).

1.3.2 Meaning and Features of data

Data refers to quantitative information that provides facts in an aggregate manner, making it useful for decision-making.

Here are some key features of statistics or data:

1. Statistics are the Aggregate of Facts

Explanation: A single fact cannot be considered statistics or data. For example, the mark secured by a single student in a mathematics exam is merely a fact, not data. However, when we collect the marks of all students in a class, it becomes data.

Example: The mark of one student in mathematics is 95. This alone is not data. But the marks of all students in the class, say

95, 85, 78, etc., represent data. Aggregate information allows for analysis and meaningful conclusions.

See the **table 1** below. It gives the marks secured by all the 18 students of a class in mathematics. By looking at this we can compare the performance of the whole class. So, this is an example of data.

Students	Marks	Students	Marks
A	95	J	35
B	90	K	30
C	75	L	85
D	65	M	20
E	90	N	90
F	100	O	80
G	80	P	70
H	45	Q	100
I	40	R	50

From the above data we can know the following

- How many students have secured more than 90?
- How many students have failed?
- How many students secured less than 50?

On the basis of the answers to these questions, the teacher can take necessary steps to improve the performance of students wherever needed. So, in this way as aggregate of facts data are more meaningful than any single information.

2. Numerically Expressed

Explanation: Statistics or data must be quantitative. Qualitative attributes like good, bad, or average cannot be considered data as their magnitudes cannot be quantified.

Example: The statement "5 students scored above 90 in mathematics" is data. Descriptive terms like "many students did well" are not data.

In the above table marks of students are given numerically. We can give another example as in **table 2** below which shows number of students admitted in the 1st year in different colleges in an imaginary city.

College	Number of Students
Govt. College	409
Savitri College	308
J.P. College	401
N.D. College	510

3. **Data are Affected by a Multiplicity of Causes**

Explanation: Data are influenced by multiple factors rather than a single cause. This complexity must be considered when analyzing data.

Example: The rise in the price of a commodity could be due to several factors such as reduced supply, increased demand, higher taxes, or increased wages.

4. **Reasonable Standard of Accuracy**

Explanation: While 100% accuracy is not always possible or necessary, a reasonable standard of accuracy is expected. This allows for practical and reliable conclusions.

Example: If a new medicine for controlling cholesterol shows a 90% success rate in clinical trials, it can be considered effective even though it is not 100% effective.

5. **Predetermined Purpose**

Explanation: Data is collected with a specific purpose in mind, which guides its collection and subsequent use.

Example: The data on students' marks in mathematics can be used to assess academic performance and identify areas for improvement. Similarly, data on the number of young people entering college can be used to gauge the level of higher education enrolment in a city.

6. **Importance of Data in Economics:** Data is crucial in various areas of economics, enabling informed decision-making and policy formulation. Here are some specific areas where data plays a significant role:

- **Economic Planning:** Utilizing historical data to forecast and plan future economic activities.
- **Determining National Income:** Collecting quantitative information to calculate the national income and assess economic health.
- **Formulating Government Policies:** Using statistical data to create policies that address economic and social issues effectively.

1.4 Classification

After the collection and editing of data, the initial step towards further processing involves classification.

Classification is the process of arranging data into sequences and groups based on their common characteristics or separating them into different classes.

Objectives:

1. **Condensation of Data:** The primary objective is to simplify and condense statistical material for easy understanding.
2. **Identification of Similarities and Differences:** Classification highlights similarities and differences among statistical facts by grouping data with similar characteristics.
3. **Facilitating Comparison:** It enables comparison by organizing data according to similarities and differences.
4. **Revealing Relationships:** Classification aids in identifying cause-and-effect relationships within the data, such as the relationship between literacy and criminal tendencies.
5. **Preparation for Tabulation:** Classification serves as the foundation for tabulation, which is crucial for statistical analysis.

1.4.1 Characteristics or Rules for a Good Classification:

1. **Unambiguity:** Classes should be clearly defined to eliminate doubts or confusion. For instance, population can be classified as literate or illiterate.
2. **Exhaustive and Mutually Exclusive:** Classification should cover all aspects (exhaustive) and ensure that each item fits into only one class (mutually exclusive). For example, students categorized as urban, rural, or hostelers should not overlap.
3. **Stability:** The classification should remain consistent over the investigation period.
4. **Suitability:** It should align with the objectives of the inquiry. For example, if studying the relationship between sex and university education, classification based on age or religion may not be necessary.
5. **Flexibility:** A good classification should be adaptable to changes in circumstances. It should easily adjust while retaining stability.

1.4.2 Methods of Classification

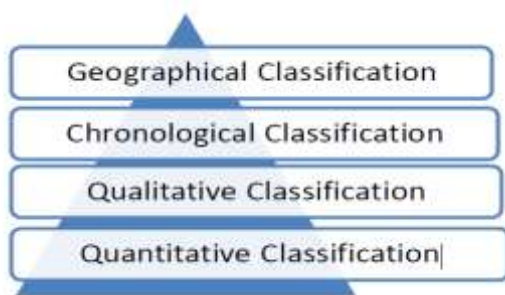


Fig3: Classification Methods.

Classification of data can be approached through various methods. Here are four primary methods of classification:

1. Geographical Classification:

- **Definition:** Data are classified based on geographic regions or locations such as countries, states, cities, or districts.
- **Purpose:** It helps in analyzing regional differences and making comparisons between different geographical areas.

Example 1: Population data divided by continents (Asia, Europe, Africa, etc.) or economic data categorized by states within a country.

Example 2: Yield of wheat in (kg/Acre) in different Countries.

Country	America	China	Denmark	France	Iraq
Yield of wheat in (kg/acre)	1925	893	225	439	862

2. Chronological Classification:

- **Definition:** Data are organized according to time periods such as years, months, weeks, or days.
- **Purpose:** This method is useful for identifying trends, patterns, and changes over time.

Example 1: Yearly sales data for a company, monthly temperature records, or daily stock market prices.

Example 2: The estimates of birth rates in India during 1970 – 76 is.

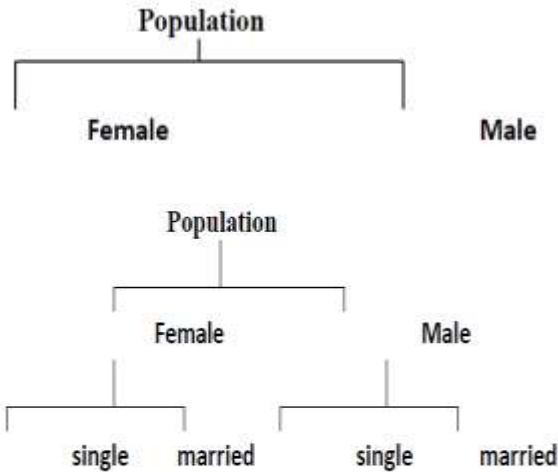
The estimates of birth rates in India during 1970 – 76 are

Year	1970	1971	1972	1973	1974	1975	1976
Birth Rate	36.8	36.9	36.6	34.6	34.5	35.2	34.2

3. Qualitative Classification:

- **Definition:** Data are categorized based on qualitative attributes or characteristics which cannot be measured numerically but can be categorized into distinct groups.
- **Purpose:** It helps in understanding the distribution of non-numerical attributes within a dataset.

Example 1 and 2 : Classification of a population based on gender (male, female), marital status (married, single), or education level (illiterate, primary, secondary, tertiary).
A simple classification may be shown as under



4. **Quantitative Classification:**

- **Definition:** Data are classified based on quantitative variables which can be measured and expressed numerically.
- **Purpose:** This method is used to analyze numerical data and identify the distribution of various numerical attributes.
-

Example 1and 2: Classification of income levels (low, medium, high), age groups (0-10, 11-20, 21-30), or height ranges (150-160 cm, 161-170 cm). 2). Weights in Kg and No. of Children’s.

Weight (in kg)	No of children
5-10	50
10-15	200
15-20	260
20-25	360
25-30	90
30-35	40
Total	1000

1.4.3 Tabulation

Definition: Tabulation is the process of systematically arranging data into rows and columns, making it easier to summarize and analyze large amounts of information.

- To present complex data in a simple, structured, and readable format.
- To facilitate comparison, identification of patterns, and effective communication of data.

Example: A table showing the number of students enrolled in different courses across various years.

1.4.4 Objectives of Tabulation

a. **To Simplify Complex Data:**

Purpose: By organizing data into rows and columns, tabulation transforms complex datasets into a more understandable and manageable format.

Benefit: It reduces the complexity of raw data, making it easier for users to interpret and draw meaningful conclusions.

b. **To Facilitate Comparison:**

Purpose: Tabulation allows for side-by-side comparison of data across different categories, time periods, or geographic regions.

Benefit: This enables easier identification of trends, patterns, and differences within the dataset.

c. **To Economize Space:**

Purpose: Tabulating data provides a compact way to present large amounts of information.

Benefit: It reduces the space required to display data compared to narrative descriptions or unstructured lists, making it more efficient and easier to handle.

d. **To Facilitate Presentation:**

Purpose: Tabulated data can be more easily incorporated into reports, presentations, and publications.

Benefit: It enhances the clarity and effectiveness of data communication, ensuring that the audience can quickly grasp the key points.

e. **To Help in Analysis of Data:**

Purpose: Tabulation is a preliminary step that organizes data systematically, setting the stage for more detailed analysis.

Benefit: It enables the application of various statistical tools and methods for deeper insights and conclusions.

f. **To Help in Reference:**

Purpose: Well-tabulated data provides an easily accessible reference point for future use.

Benefit: It allows for quick retrieval and verification of data, making it a valuable resource for ongoing research or decision-making processes.

1.4.5 Difference Between Classification and Tabulation

Basis of Difference	Classification	Tabulation
Presentation	Classifies data into different classes based on characteristics.	Organizes data into rows and columns.
Sequence	First step in data processing.	Second step after classification.
Methods	A method of statistical analysis used to categorize data.	A method of data presentation used to systematically arrange data.
Use of Data	Uses original, unprocessed data.	Can use derived data such as percentages, coefficients, and proportions.

1.5 Presentation of Data

Data collected in the form of schedules and questionnaires are not self-explanatory. These are in the form of raw data. In order to make them meaningful, these are to be made presentable Classification and tabulation are the basic tools of presenting raw data in systematic way.

1.5.1 Tabular Presentation of Data

Tabulation of data may be defined as the logical and systematic organisation of statistical data in rows and columns, designed to simplify the presentation and to facilitate quick comparison. In tabular presentation errors and emissions could be readily detected. Another advantage of tabular presentation is avoidance of repetition of explanatory terms and phrases.

A table constructed for presenting the data has the following parts:

- **Title:** This is brief description of the contents and is shown on the top of the table.
- **Stub:** The extreme left part of a table is called Stub. Here the descriptions of the rows are shown.
- **Caption and Box Head:** The upper part of the table, which shows the description of columns and sub columns, is called

Caption. The row of the upper part, including caption, units of measurement and column number, if any, is called box-head.

- **Body:** This part of the table shows the figures.
- **Footnote:** In this part we show the source of data and explanations, if any.

The diagram illustrates the structure of a table. A horizontal dashed arrow at the top is labeled "Title". A vertical dashed arrow on the left is labeled "S T U B". The table itself has a box-head with two rows. The first row of the box-head contains cells labeled (1), (2), (3), (4), (5), (6), and (7). The second row of the box-head contains cells labeled (2), (3), (5), and (6). The body of the table consists of a large empty rectangular area. At the bottom of the table is a single row of empty cells, representing the footnote section.

Title						
(1)	(2)	(3)	(4)	(5)	(6)	(7)

- 1.6. Distributions:** organizing and analyzing data is called **Distributions**. To extract meaningful information from data, it is often necessary to organize it using **frequency distributions**. This involves grouping data into categories, or classes, and counting the number of observations that fall into each class.

1.6.1 Frequency Distribution

Frequency distribution is a tabular arrangement of data that shows the frequency of each item. According to Croxton and Cowden, "Frequency distribution is a statistical table in which different values of a variable are shown in the sequence of magnitude along with corresponding frequencies."

1.6.2 Types of Frequency Distribution

1. **Discrete Frequency Distribution:** This is a discontinuous frequency distribution where observations are independent of each other. Each observation is distinct and separate from others.

Example: The number of children in 20 families: 1, 1, 2, 3, 4, 3, 2, 1, 1, 4, 5, 2, 4, 2, 2, 1, 3, 3, 2, 5.

Construction of Discrete Frequency Distribution:

- Prepare a table with three columns.
- In the 1st column, list all possible values of the variable.
- In the 2nd column, put tally marks or tally bars, keeping track of the repetition of each value.
- Count the tally marks and record this count in the 3rd column, titled 'frequency'.

No. of children	Tally Bars	Frequency
1		5
2		6
3		4
4		3
5		2

2. **Continuous Frequency Distribution:** This type of distribution involves data arranged in classes or groups that are not exactly measurable. The class intervals are continuous, covering the entire range of data.

Example: 39, 25, 5, 33, 19, 21, 12, 48, 13, 21, 9, 1, 10, 9, 8, 12, 17, 40, 12, 46, 37, 17, 27, 30, 6, 2, 23.

Construction of Continuous Frequency Distribution:

- **Find out the range:**
 - $\text{Range} = \text{Maximum value} - \text{Minimum value}.$
- **Class size:** *Decide the class size based on the range and the number of classes you want.*
- **Width of class interval:**
 - $\text{Width of class interval} = \text{Range} / \text{Number of classes}.$
- **Mid value:**
 - $\text{Mid value of a class interval} = \text{Lower limit of class} + \text{Upper limit of class} / 2$

- **Class frequency:** *The number of observations corresponding to a particular class is known as the frequency of the class.*

Marks	Tally Bars	No. of Students
0 – 10		7
10 – 20		8
20 – 30		5
30 – 40		4
40 – 50		3

Types of Series According to Class Interval:

- Inclusive Form:** A frequency distribution where the upper limit of each class is included.
Example: 0-9, 10-19, 20-29, etc.
- Exclusive Form:** A frequency distribution where the upper limit of a class is excluded and included in the next class.
Example: 0-10, 10-20, 20-30, etc.

1.6.3 Graphical Representation of Distributions:

Charts and diagrams are useful devices for the data presentation. Diagrams are appealing to the eyes and they are helpful in assimilating data readily and quickly. A chart, on the other hand, can clarify complex problems and reveal the hidden facts. But charts or diagrams unlike tables do not show details of data and require much time to construct. Note that the words 'charts' and 'diagrams' are used almost in the same sense. The common types of charts and diagrams are,

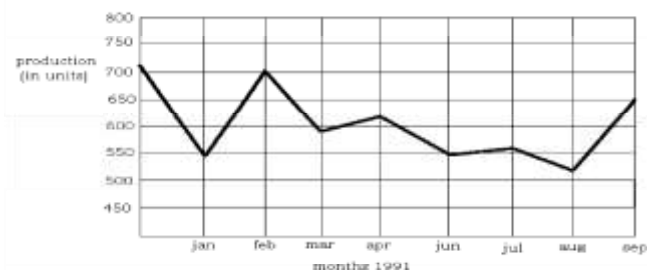
Charts and diagrams are powerful tools for data presentation, providing visual appeal and facilitating quick comprehension of information. While charts and diagrams may not provide the level of detail found in tables and can be time-consuming to create, they offer a clear and often compelling way to communicate complex data.

Below are descriptions of common types of charts and diagrams:

- Line Diagrams:** Line diagrams, or line graphs, are particularly useful for displaying time series data and showing trends over time. They are widely used in business and commerce to illustrate the relationship between two variables.
 - Construction:**
 - Axes:** A horizontal axis (X-axis) and a vertical axis (Y-axis) intersect at the origin.

- **Data Points:** Data points are plotted on the graph paper, representing values at specific intervals.
- **Line:** Points are connected by straight lines or curves to form the diagram.
- **Types:**
 - **Natural Scale:** Equal distances represent equal amounts of change.
 - **Ratio Scale:** Equal distances represent equal ratios.

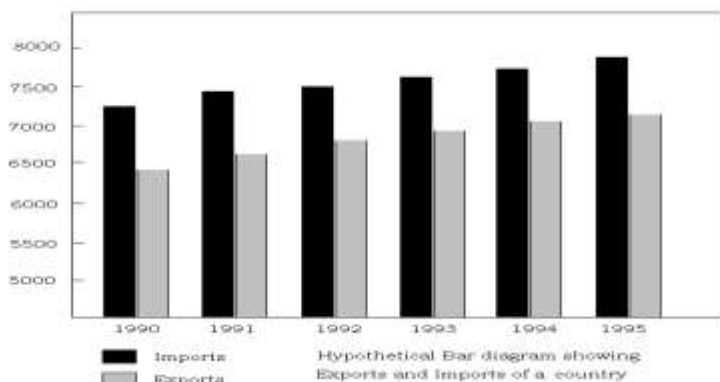
Example: A line diagram could show the sales figures of a company over several months, with time on the X-axis and sales figures on the Y-axis.



2. **Bar Diagrams:** Bar diagrams use rectangular bars to represent different categories of data. They are useful for comparing the frequency, count, or other measures across categories.

- **Construction:**
 - **Bars:** Equally spaced rectangular bars are drawn for each category.
 - **Baseline:** Bars start from a common baseline and have equal width.
 - **Length:** The length of each bar represents the value of the data.
- **Types:**
 - **Vertical and Horizontal Bars:** Depending on orientation.
 - **Grouped Bar Diagrams:** Compare two or more sets of related data.
 - **Subdivided Bar Diagrams:** Compare the sizes of different components within each bar.
 - **Paired Bar Diagrams:** Consist of several pairs of horizontal bars for comparative purposes.

Example: A bar diagram could show the number of houses sold in different zip codes, with zip codes on the X-axis and the number of houses on the Y-axis.

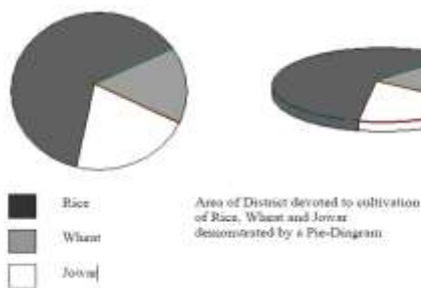


3. **Pie Diagrams:** Pie diagrams, or pie charts, display data as slices of a circle. Each slice represents a category's proportion of the total.

- **Construction:**

- **Circle:** The entire circle represents the total data set.
- **Slices:** The circle is divided into slices where each slice's angle (in degrees) represents the category's percentage of the total (percentage multiplied by 3.6 degrees).

Example: A pie diagram could show the market share of different smartphone brands, with each slice representing the percentage share of each brand.

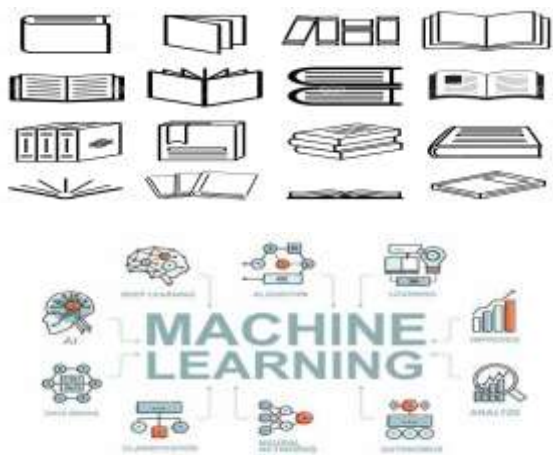


4. **Pictogram:** Pictograms use images or symbols to represent data. Each image or symbol corresponds to a specific quantity, making it easy to visualize data.

- **Construction:**

- **Images/Symbols:** Rows of pictures or symbols of equal size represent quantities.
- **Numerical Value:** Each picture or symbol corresponds to a definite numerical value.

Example 1 and 2: A pictogram might represent the number of books sold using small book icons, with each icon representing books.



Summary Table:

Chart/Diagram	Type	Best For	Key Characteristics
Line Diagram	Time Series Data	Trends over time	Plots data points connected by lines on a graph
Bar Diagram	Categorical Data	Comparing categories	Uses rectangular bars with equal width
Pie Diagram	Proportional Data	Showing parts of a whole	Circle divided into proportional slices
Pictogram	Simplified Quantitative Data	Illustrating data with symbols	Uses pictures or symbols to represent quantities

Sample 1: Grouped Frequency Distribution of Age in a Locality.

Class Interval	Class frequency	Class limit		Class boundaries		Class mark	Width of Class	Frequency Density	Relative Frequency
		lower	upper	lower	upper				
15 – 19	37	15	19	14.5	19.5	17	5	7.4	0.185
20 – 24	81	20	24	19.5	24.5	22	5	16.2	0.405
25 – 29	43	25	29	24.5	29.5	27	5	8.6	0.215
30 – 34	24	30	34	29.5	34.5	32	5	4.8	0.12
35 – 44	9	35	44	34.5	44.5	39.5	10	0.9	0.045
45 – 59	6	45	59	44.5	59.5	52	15	0.4	0.03

In the example of grouped frequency distribution, we have shown few useful terms associated with it. They are class intervals or class, class frequency, cumulative frequency (greater than and less than type), class limits (upper and lower), class boundaries (upper and lower), midpoint of class interval, width of a class, relative frequency and lastly frequency density. We will formally define these terms.

- ✓ **Class:** When dealing with a large set of observations that vary widely, data are often grouped into several categories based on the size of their values. Each of these categories is known as a class interval or simply a class.
- ✓ **Class Frequency:** This is the number of observations that fall within each class interval. It is often referred to simply as frequency.
- ✓ **Class Limits:** These are the two numbers that define the range of a class interval. They are used to tally the original observations into their respective classes.
- ✓ **Class Boundaries:** These are the extreme values that could ever be included in a class interval. They mark the outermost limits of each class.
- ✓ **Mid-Point of Class Interval:** Also known as the class mark or mid-value, this is the value located exactly in the middle of a class interval. It serves as the representative value of that interval and is calculated as:

$$\text{Mid-point of Class Interval} = (\text{Lower Class Boundary} + \text{Upper Class Boundary}) / 2$$

- ✓ **Width of a Class:** This is the difference between the upper- and lower-class boundaries and defines the span of the class interval. It is calculated as:

$$\text{Width of a Class} = \text{Upper Class Boundary} - \text{Lower Class Boundary}$$

- ✓ **Relative Frequency:** The relative frequency of a class is the proportion of the total number of observations that fall within that class. It is calculated as:

$$\text{Relative Frequency} = \text{Class Frequency} / \text{Total Frequency}$$

- ✓ **Frequency Density:** This is the frequency of a class per unit width of the class interval. It is calculated as:

$$\text{Frequency Density} = \text{Class Frequency} / \text{Width of the Class}$$

- ✓ **Cumulative Frequency:** This refers to the number of observations that are less than or equal to (or greater than) a certain value or class interval.

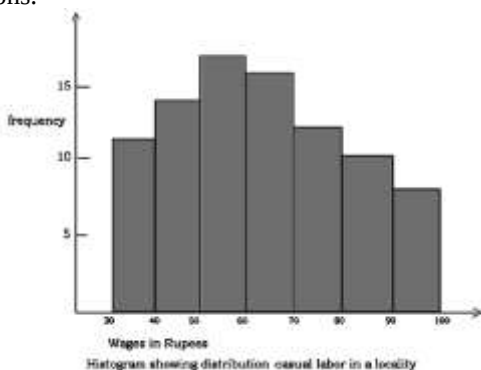
- ✓ **Less-than type cumulative frequency distribution:** The number of observations up to a given value or class.
- ✓ **More-than type cumulative frequency distribution:** The number of observations greater than a given value or class.

1.6.4 Histogram, Frequency Polygon and Ogives

Histograms, frequency polygons, and ogives are graphical tools used to illustrate frequency distributions. Here's a concise explanation of each:

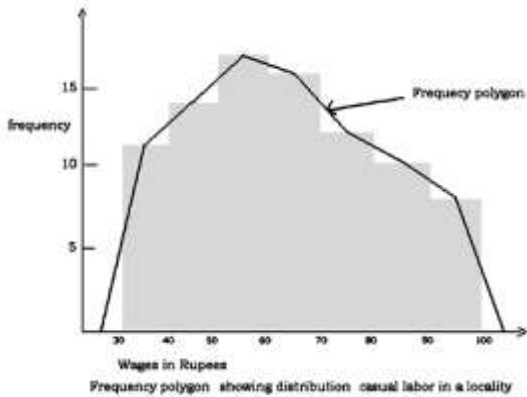
1. Histogram:

- **Description:** Histograms are commonly used for presenting grouped frequency data. They consist of adjacent rectangles aligned along a common baseline.
- **Construction:** The width of each rectangle represents the class width, while the height indicates the frequency density.
- **Purpose:** Histograms provide a visual depiction of the distribution of data, making it easy to discern patterns and variations.



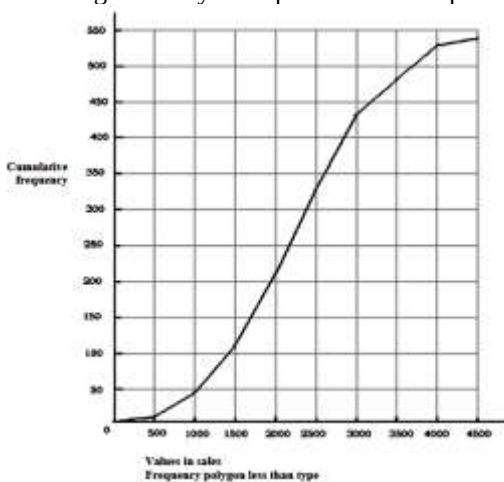
2. Frequency Polygon:

- **Description:** Frequency polygons are created by connecting the midpoints of the tops of consecutive rectangles in a histogram.
- **Construction:** Line segments join the midpoints, forming a smooth curve. The endpoints connect to the baseline at the mid-values of any empty classes.
- **Purpose:** Frequency polygons offer a smoother alternative to histograms, aiding in the visualization and comparison of data distributions.



3. Ogives (Cumulative Frequency Graphs):

- **Description:** Ogives represent cumulative frequency distributions graphically.
- **Construction:** Plotting cumulative frequencies against the mid-values of class intervals and connecting them with line segments produces ogives.
- **Purpose:** Ogives help visualize cumulative distributions, providing insights into cumulative frequencies and facilitating the analysis of percentiles and quartiles.



Histograms depict grouped frequency data using adjacent rectangles, frequency polygons connect midpoints to form smooth curves, and ogives represent cumulative distributions graphically. Each of these tools aids in the visual representation and understanding of frequency distributions in data analysis.

Example: Following data were obtained from a survey on the value of annual sales of 534 firms. Draw the histogram and the frequency polygon and ogive from the data.

Value of sales	Number of firms
0-500	3
500-1000	42
1000-1500	63
1500-2000	105
2000-2500	120
2500-3000	99
3000-3500	51
3500-4000	47
4000-4500	4

Solution:

1.6.5 Stem and Leaf chart:

A stem-and-leaf plot is a method of organizing and displaying numerical data. It provides a way to visualize the distribution of data while retaining individual data points. Here's an explanation of how it works:

Stem-and-Leaf Plot:

- **Structure:** In a stem-and-leaf plot, each data point is divided into two parts: the stem and the leaf. The stem represents the leading digits of the data, while the leaf represents the trailing digits.
- **Stem:** The stem consists of the leftmost digits of the data. It forms the vertical axis of the plot.
- **Leaf:** The leaf consists of the rightmost digit of the data. It is placed next to the corresponding stem to form the plot.
- **Organization:** The data points are sorted numerically, and the stems are arranged in ascending order from top to bottom. The leaves associated with each stem are then listed in ascending order next to the corresponding stem.

Advantages:

1. **Retains Individual Data Points:** Unlike other graphical representations like histograms or box plots, a stem-and-leaf plot retains the actual data points, making it easier to identify specific values.
2. **Quick Visualization:** Stem-and-leaf plots provide a quick and intuitive way to visualize the distribution of data, especially for small to moderate-sized datasets.

3. **Easy Comparison:** Comparing multiple datasets or groups is straightforward with stem-and-leaf plots, as they can be displayed side by side for easy comparison.

Limitations:

1. **Limited for Large Datasets:** Stem-and-leaf plots can become cumbersome for large datasets, as they may result in many stems and leaves, making interpretation difficult.
2. **Less Detailed than Histograms:** While stem-and-leaf plots provide a visual representation of the data, they may lack the level of detail offered by histograms or other graphical methods for larger datasets.

Example: Suppose we have the following dataset:
23, 36, 47, 51, 59, 63, 68, 74, 82, 94.

We can construct a stem-and-leaf plot as follows:

Stem	Leaf
2	3
3	6
4	7
5	1 9
6	3 8
7	4
8	2
9	4

Interpretation: The stem-and-leaf plot allows us to see the distribution of the data and identify any patterns or outliers. In the example above, we can see that most of the data points are clustered around the middle (stem 5 and 6), with a few outliers on either end.

1.6.6 Statistics and the Data Analysis Process

Statistics involves three critical tasks:

1. Collecting data.
2. Summarizing data.
3. Analyzing data.

Importance of Each Task:

- Without summarization and analysis, raw data are of little value.
- Sophisticated analyses require data collected sensibly.

Purpose of Statistical Studies:

- Answer questions about our world.

Examples:

- Effectiveness of a new flu vaccine.
- Trends in bicycle helmet use.
- Severity of injuries with and without helmets.
- Number of credit cards college students have.
- Textbook costs for engineering vs. psychology students.

1.6.7 Data Analysis Process:

1. Understanding the Problem:

- Know the research goal and the questions to be answered.
- Have a clear direction before gathering data.

2. Deciding What and How to Measure:

- Determine the necessary information to answer the questions.
- Define variables clearly and choose appropriate measurement methods.

3. Data Collection:

- Decide if existing data is sufficient or if new data is needed.
- Understand the source and purpose of existing data.
- Plan carefully if new data is to be collected.

4. Data Summarization and Preliminary Analysis:

- Summarize data graphically and numerically.
- Gain insights into data characteristics and guide further analysis.

5. Formal Data Analysis:

- Select and apply appropriate statistical methods.
- Use methods taught in statistics courses or textbooks.

6. Interpretation of Results:

- Determine what can be learned from the data.
- Draw conclusions and guide future research.
- Good data analysis is iterative, often leading to new research questions.

Note:**Statistics is Required Prerequisite**

Machine learning and statistics are two tightly related fields of study. So much so that statisticians refer to machine learning as applied statistics or statistical learning rather than the computer-science-centric name. Machine learning is almost universally presented to beginners assuming that the reader has some background in statistics. We can make this concrete with a few cherry-picked examples.

Take a look at this quote from the beginning of a popular applied machine learning book titled Applied Predictive Modeling:

... the reader should have some knowledge of basic statistics, including variance, correlation, simple linear regression, and basic hypothesis testing (e.g., p-values and test statistics).

| Page vii, *Applied Predictive Modeling*, 2013.

Here's another example from the popular Introduction to Statistical Learning book:

We expect that the reader will have had at least one elementary course in statistics.

| Page 9, *An Introduction to Statistical Learning with Applications in R*, 2013.

Even when statistics is not a prerequisite, some primitive prior knowledge is required as can be seen in this quote from the widely read Programming Collective Intelligence:

... this book does not assume you have any prior knowledge of [...] or statistics. [...] but having some knowledge of trigonometry and basic statistics will help you understand the algorithms.

| Page xiii, *Programming Collective Intelligence: Building Smart Web 2.0 Applications*, 2007.

In order to be able to understand machine learning, some basic understanding of statistics is required. To see why this is the case, we must first understand why we need the field of statistics in the first place.

Statistics in Plural Sense.

Statistics, in the plural sense, refers to numerical data like population and employment statistics, but any numerical information is not considered statistical.

Example: Ram gets ₹ 100 per month as pocket allowance is not Statistics. It is neither an aggregate nor an average. Whereas average pocket allowance of the students of Class X is ₹100 per month and there are 80 students in class XI & 8 students in Class XII are Statistics.

The following table shows a set of data that which is Statistics and which is not Statistics.

<i>Data which are not Statistics</i>	<i>Data which are Statistics</i>
<i>A cow has 4 legs. Average height of the 26 plus male people in</i>	<i>A cow has 4 legs. Average height of the 26 plus male people in</i>
<i>India is 6 feet compare to 5 feet in Nepal.</i>	<i>India is 6 feet compare to 5 feet in Nepal.</i>
<i>Ram has 200 rupees in his pocket. Birth rate in India is 18 per thousand compares</i>	<i>Ram has 200 rupees in his pocket. Birth rate in India is 18 per thousand compares</i>

From above information we can say that “All statistics are data, but all data are not statistics”.